

A New Role for RNA G-quadruplexes in Aging and Alzheimer's Disease

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eLife Assessment

The current human tissue-based study provides **convincing** evidence correlating hippocampal expressions of RNA guanine-rich G-quadruplexes with aging and with Alzheimer's Disease presence and severity. The results are **important** and hold promise for deeper understanding of AD's pathogenesis and potential new therapeutic strategies.

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Abstract

Introduction

As the world population ages, new molecular targets in aging and Alzheimer's Disease (AD) are needed to combat the expected influx of new AD cases. Until now, the role of RNA structure in aging and neurodegeneration has largely remained unexplored. **METHODS:** In this study, we examined human hippocampal *postmortem* tissue for the formation of RNA G-quadruplexes (rG4s) in aging and AD.

Results

We found that rG4 immunostaining strongly increased in the hippocampus with both age and with AD severity. We further found that neurons with accumulation of phospho-tau immunostaining contained rG4s, that rG4 structure can drive tau aggregation, and that rG4 staining density depended on APOE genotype in the human tissue examined.

Discussion

Combined with previous studies showing the dependence of rG4 structure on stress and the extreme power of rG4s at oligomerizing proteins, we propose a model of neurodegeneration in which chronic rG4 formation drives proteostasis collapse. These morphological findings suggest that further investigation of RNA structure in neurodegeneration is a critical avenue for future treatments and diagnoses.

1. Introduction

As the aged population increases, Alzheimer's Disease (AD) and AD-related dementias (ADRDs) will become one of the world's largest medical and economic crises due to the lack of early diagnostic methods and durable therapies^{1,2}. Age is the highest risk factor for AD, as disease prevalence increases exponentially as a function of age, and close to 40% of older adults over the age of 80 experience AD or ADRDs^{3,4}. The most important genetic risk factor for developing late onset AD (LOAD) is the Apolipoprotein E (APOE) genotype, at least in Caucasians, where carriers of one or two copies of the APOE4 allele have a substantial increase in the risk for developing LOAD when compared to carriers of the APOE2 or 3 alleles^{4,5}. Beyond APOE, recent meta-analyses combining genome-wide association studies (GWAS) have expanded the number of AD-risk loci⁶, but most of the disease-associated variants reside in non-protein coding regions of the genome, making it difficult to elucidate how they affect AD susceptibility and pathogenesis.

The exact mechanisms behind these two predominant AD risk factors remains a mystery but both are widely associated with increased abnormal protein aggregation, which is the most abundantly observed pathological hallmark of AD pathogenesis. AD and ADRDs all have an abnormal protein aggregation profile that includes: 1) extracellular amyloid senile plaque deposition occurring at least 5-10 years before mild cognitive impairments (MCI) are observed and, 2) aggregates of phosphorylated Tau (p-Tau) forming intracellular neurofibrillary tangles (NFTs) near MCI onset^{7,8}. Novel biomarker studies have shown that hyperphosphorylated Tau species exist in plasma prior to onset of dementia symptoms, with specific phosphorylation sites seeming to affect the affinity of binding of the Tau protein to microtubules⁹. With advancing age, protein aggregation is accelerated by the accumulation of reactive oxygen species (ROS) as well as the onset of neuroinflammation in the brain¹⁰⁻¹². Having APOE4 was shown to significantly increase accumulation of amyloid and NFTs in Caucasians, and also increases inflammation and reactive oxygen species in the human brain^{5,13}. The effects of these AD risk factors have been studied mostly at the protein level in the brain, but not yet at RNA level, where structure and function have a myriad of important roles in numerous biological processes.

RNA higher order structures of particular interest are RNA guanine-rich G-quadruplexes (rG4s)¹⁴. G-quadruplexes are secondary structures that can fold under physiological conditions by guanine-rich DNA and RNA molecules and occur in the genomes and transcriptomes of many species, including humans¹⁵. rG4s are four-stranded helices formed by guanine tetrads through Hoogsteen base-pairing, with a monovalent cation channel running down the middle of the tetrads (**Fig. 1a**)¹⁶. G4s are potentially involved in the regulation of DNA and RNA processes, such as replication, transcription, translation, RNA localization, and degradation¹⁷⁻¹⁹. Recently, rG4s were shown to form as a function of stress in eukaryotic cells²⁰. Under non-stress conditions, rG4s were largely unfolded, but upon introduction of stress (including ROS stress) the rG4s preferentially folded and remained folded until the cessation of stress²¹. This observation raises the possibility that rG4s could preferentially fold under the increased and chronic stress conditions of aging and AD in the human brain, leading to abnormally high levels of rG4

formation in older adults or in adults with protein misfolding diseases²². RNA has been known to be included in AD-related aggregates of both amyloid and NFTs in patients with AD for decades²³, and a recent sequencing study of nucleic acids embedded in AD aggregates showed an enrichment in potential rG4-forming sequences²⁴.

It has recently been shown that rG4s can greatly enhance protein oligomerization²⁵, ²⁶. Moreover, rG4s can bind to and greatly enhance the phase transition of tau²⁷. Similarly, Zwierzchowski-Zarate et al explored the effects of different RNA sequences on tau aggregation in a cellular model²⁸. In their study, several sequences caused an increase in tau aggregation, but the highest level of tau aggregation was achieved with a GGGC repeat sequence²⁸. However, the structure of the RNAs, including possible rG4 formation, was not explored or speculated upon. We therefore performed CD spectroscopy to determine if the sequence associated with the highest levels of tau fibrillation might be forming rG4 secondary structures. The CD spectra show that this RNA forms parallel rG4 structures *in vitro* (Supplementary Data Fig. 1).

Based on these previous *in vitro* and cellular studies, an increased presence of rG4s in the brain could contribute to AD-associated aggregation. However, the presence of rG4s in the human brain or other human tissues has not been shown previously, nor how it relates to aging or AD pathology. This was the focus of the current study.

2. Methods

Human brain tissue

This work was approved by the University of Colorado Institutional Review Board (IRB) and the study was performed in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. However, the use of *postmortem* brain tissue for research is exempt according to federal IRB regulations and does not require IRB review and approval. All efforts were made to consider diversity, equity and inclusion (DEI) in the selection of brain tissues for this study. The tissues included in this study were obtained from the Medical University of South Carolina (MUSC) brain bank (Dr. Eric Hamlett). *Postmortem* consent was obtained from the next of kin, and the brain was rapidly removed. **Fig. 1b** shows the demographics for the 21 cases used for this study. The tissue was sliced in 1 cm coronal slices and subjected to free-floating fixation in 4% paraformaldehyde in phosphate buffered saline (PBS) for 72 hours. The fixed tissue was then transferred to a 30% sucrose in phosphate buffer (PB), followed by transfer to a cryoprotection solution in PBS (30% Glycerol, 30% Ethylene Glycol and 40% PO₄ Buffer) and stored in -20° C until dissection. Anatomical blocks from 19 brain regions were dissected from the 1 cm slabs, embedded in paraffin and cut at a thickness of 5µm on a Microm Microtome (Thermo Fisher Scientific Inc., Waltham, MA) and stained for routine neuropathological staging according to the NIA-AA Revised Criteria for Diagnosis and Staging of Alzheimer's Disease²⁹.

Immunofluorescence

Paraffin-embedded sections were deparaffinized in xylene (RPI, CAT#111056) and rehydrated using wash steps at: 100%, 95%, 70%, 50% ethanol followed by distilled water. Heat-induced antigen retrieval was performed using 0.05% citraconic acid at pH 7.2 for 20 minutes in a steamer. Slides were then washed with a saline solution buffered with tris(hydroxymethyl) aminomethane (Tris, TBS) and blocked with 0.25% triton-X and 5% normal serum for 1 hour. After blocking, slides were rinsed again with TBS and incubated with 1X True Black Lipofuscin Autofluorescence Quencher (Biotium CAT#23007) diluted in 70% ethanol. Slides were washed 3 times with TBS then incubated with primary antibodies listed in Supplementary Data Table 1 overnight, followed by Goat anti-mouse AlexaFluor 555 (Thermo Scientific #A32727) or Donkey anti-rabbit Alexafluor 488 (Invitrogen #A-21206) at 1:500 dilution for 1 hour. Slides were then washed 3 times with TBS and cover-slipped using mounting media containing DAPI (Thermo Scientific, CAT#P36931). Staining

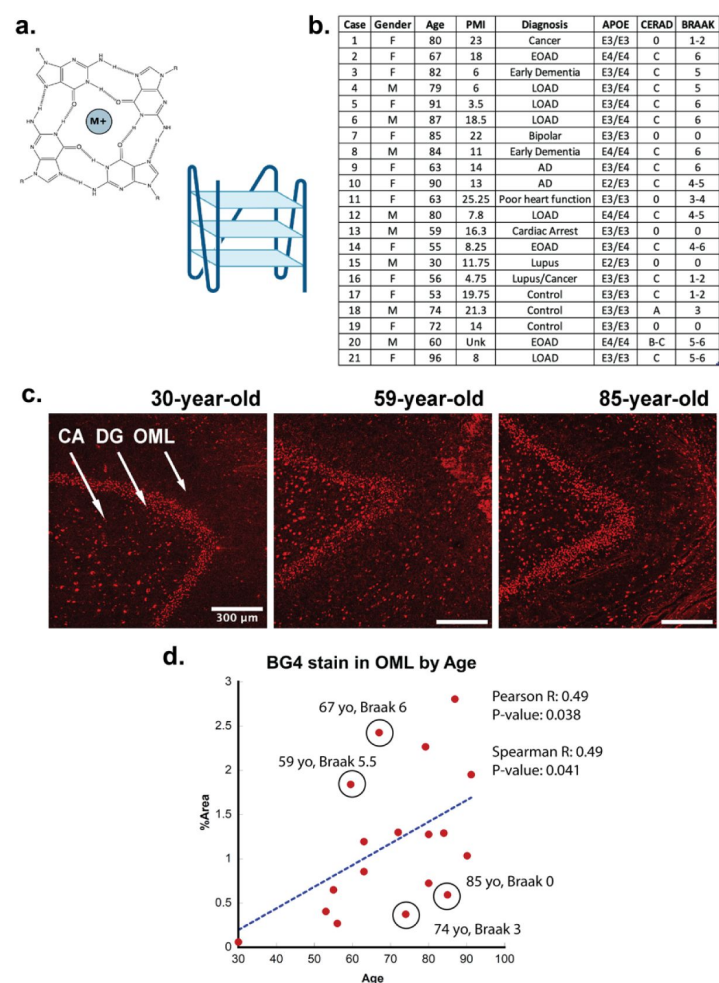


Fig 1.

a. (left) Representative G-quadruplex tetrad structure with Hoogsteen base pairing stabilized by a cation at the center. (right) G-quadruplex structure with stacked tetrads. **b.** Demographics table for individuals included in this study. **c.** Immunofluorescence showing BG4 staining (red) in brain sections of human hippocampus from four control individuals of increasing age ranging 30-91 years old (Case 15, 13, and 8, respectively) all with Braak stage 0. **d.** Correlation of percent area covered by BG4 fluorescence in the outer-molecular layer versus age of individual.

controls included omission of primary or secondary antibodies (Supplementary Data Fig. 2). In **Fig. 3b**, we performed sequential double labeling, first with BG4, and then with rRNA antibody at 1:250 (Abcam #ab17119).

In addition, we examined whether Braak stage correlated with aging itself (Supplementary Data Fig. 3). Although not significant at the 0.05 level, there was a weak correlation between Braak and Age. Braak staging is a paradigm used to describe density and regional spread of neurofibrillary tangles (NFTs) in the Alzheimer brain and includes levels 0-VI. CERAD (Consortium to Establish a Registry for Alzheimer's Disease) is a staging paradigm for Alzheimer's disease that takes into account amyloid plaque formation.

Imaging and Statistical methods

Slides were imaged on a confocal microscope at 10X, 20X and 60X magnification at wavelengths 355, 488, and 555. Images used for quantification were taken at constant microscope and intensity settings. All image quantification was performed using Fiji ImageJ. Quantification was performed by a researcher who was blinded to the neuropathological diagnosis of individual cases and with no knowledge of Braak Stage or Age for the case being quantified. Supplemental Table 2 presents side by side comparisons of quantification from 2 researchers who were both blinded to group diagnosis and individual cases. Data used in the paper are highlighted. One data point with too much variation between quantifications was removed to increase rigor. Background was subtracted using a rolling ball radius. Images were then divided into 3 regions of interests (ROI); OML, DG and CA4. (Supplementary Data Fig. 4) Images were submitted to an auto-threshold and particle's analyzed to get a percent area measurement of BG4 staining. These were graphed and linearly fit using Kaleidagraph using Pearson fit. Spearman and Pearson R and p-values are reported for 18 cases for age correlation and 18 cases for Braak correlation. An unpaired T-test with unequal variance was performed for the two groups E2/E3 and E3/E3 vs E3/E4 and E4/E4. When the singular African American case was removed from the APOE quantification, the t-probability remained highly significant at 0.0045 vs 0.002. All quantification was performed on 10X images obtained from an Olympus microscope.

Circular Dichroism

CD spectra were obtained using a Jasco J-1100 circular dichroism at 25°C. Sequence was resuspended in 10mM potassium phosphate pH 7.5 buffer and diluted to 12.5 μ M (per strand) RNA. The CD measurement was taken from 300 nm to 210 nm at 1nm intervals using a 50 nm/min scanning speed. The sequence used was 5'-CGGGCGGGGGGGGGCCCGGGCGGGGGGGCCCGGGCG-3'. Parallel structure was determined using previously described quantification methods for quadruplex topology.

3. Results

3.1 Formation of rG4s in aging

To determine if rG4s form in the human brain, we used the well-characterized single-chain BG4 anti-quadruplex antibody (Sigma-Aldrich MABE917, antibody null controls in Supplementary Data Fig. 2) to stain human hippocampal *postmortem* tissue of different ages under conditions that strongly favor cytoplasmic rG4 identification using triton-X fixation in cells. The tissues were obtained in collaboration with the brain bank at the Medical University of South Carolina (MUSC). The study included 21 cases, male and female, of different ages and with or without a clinical or neuropathological diagnosis of AD (see **Fig. 1b**). All cases had received a neuropathological examination including assessment of Braak and CERAD stages according to the updated NIA criteria except in those cases in which no pathology was noted.

First, the rG4 staining profiles were compared between different ages, with an age range of 30-92 years of age (**Fig. 1c**). We discovered a significant age-related increase in rG4 immunostaining, with the densest staining observed in older individuals (**Fig. 1c,d**). The largest difference in staining between young and older cases was observed in the outer molecular layer (OML) and in the hilar region, CA4 (see **Fig. 1c**, legend arrows). The strongest overall staining for BG4 was found in the dentate gyrus granular cell layer (DG), but with less observable age-related increases. Staining density measurements confirmed these findings and demonstrated a highly significant increase in BG4 staining in the oldest cases studied within the OML (**Fig. 1d**) and the CA4 regions of cornu ammonis (Supplementary data Fig. 5a), confirming increased intracellular BG4 accumulation as a function of age. Plotting BG4 percent area versus age confirmed the observed findings and showed correlation (Spearman $R = 0.49$) with high significance ($p = 0.038$ in the OML region (**Fig. 1d**)). Positive and significant Pearson but not Spearman correlations were observed between age and BG4 area in the CA4 region, Supplementary data Fig. 5a), and non-significant correlations were observed in the dentate granule cell layer. No significant correlation was found with BG4 stain and *postmortem* interval (PMI) (Spearman $R = -0.18$, $p = 0.49$ see Supplementary Data Fig. 6). In total, these data show that rG4s increased as a function of age in the human brain. We observed multiple outliers in both older and younger individuals due to Braak stage. For example, an 85-year-old individual with Braak stage 0, or 59-year-old individual with Braak stage 5.5 were outliers as a function of age. This led us to analyze whether rG4 formation increased with Braak stage, a measure of AD severity.

3.2 Formation of rG4s in AD

Next, we examined whether the formation of rG4s was associated with AD neuropathology. Braak staging is a neuropathological staging method that takes into account the density of neurofibrillary tangles (NFTs) both within each brain region and also from region to region³⁰, with higher numerical values representing greater AD severity⁷. Like in the case of aging, increased BG4 staining was readily apparent in AD cases, especially in those cases associated with a higher Braak stage (**Fig. 2a**). In the OML, plotting BG4 percent area versus Braak stage demonstrated a strong correlation (Spearman $R = 0.72$) with highly significantly increased BG4 staining with higher Braak stages ($p = 0.00086$) (**Fig. 2b**). Positive significant correlations ($R = 0.52$, $p = 0.028$) between Braak stage severity and BG4 staining were also observed within the CA4 region (Supplementary data fig. 4). Interestingly, the OML region is highly involved in AD pathology, and is a hippocampal region where AD pathology is often observed,³⁵ suggesting that the presence of rG4s in this particular brain region might have significance for amyloid as well as NFT formation during the AD disease process. In addition, a prominent loss of synapses in AD is found in the OML, suggesting a potential close connection with layer 2 of the entorhinal cortex³⁶.

If rG4 formation is more prevalent in AD than in non-AD cases and directly involved in AD pathology, we would also predict that rG4 formation would be more prevalent in APOE genotypes that are predisposed for higher LOAD risk, i.e. patients with one or two copies of the APOE4 genotype. Indeed, there was a significant (t-probability 0.002) increase in BG4 staining levels in the OML in *postmortem* cases with one or more APOE4 alleles compared to those with only APOE2 or APOE3 alleles (**Fig. 2c**). This result, although correlative, is consistent with a previous study performed *in vitro* in which the RNA sequences found in aggregates in different APOE genotype cell lines were different²⁴.

3.3 Patterns of rG4 localization

Images obtained at high magnification using a confocal microscope (Olympus) showed that a large amount of the BG4 immunostaining was localized in the cytoplasm near the nucleus in a punctate pattern, reminiscent of known rG4 formation *in vitro* and in cell lines in rRNA ribosomal extensions³³. We therefore also stained with an anti-rRNA antibody to determine whether this was the case in human tissue as well. As expected, the stains overlapped (**Fig. 3b**, Fig. S7),

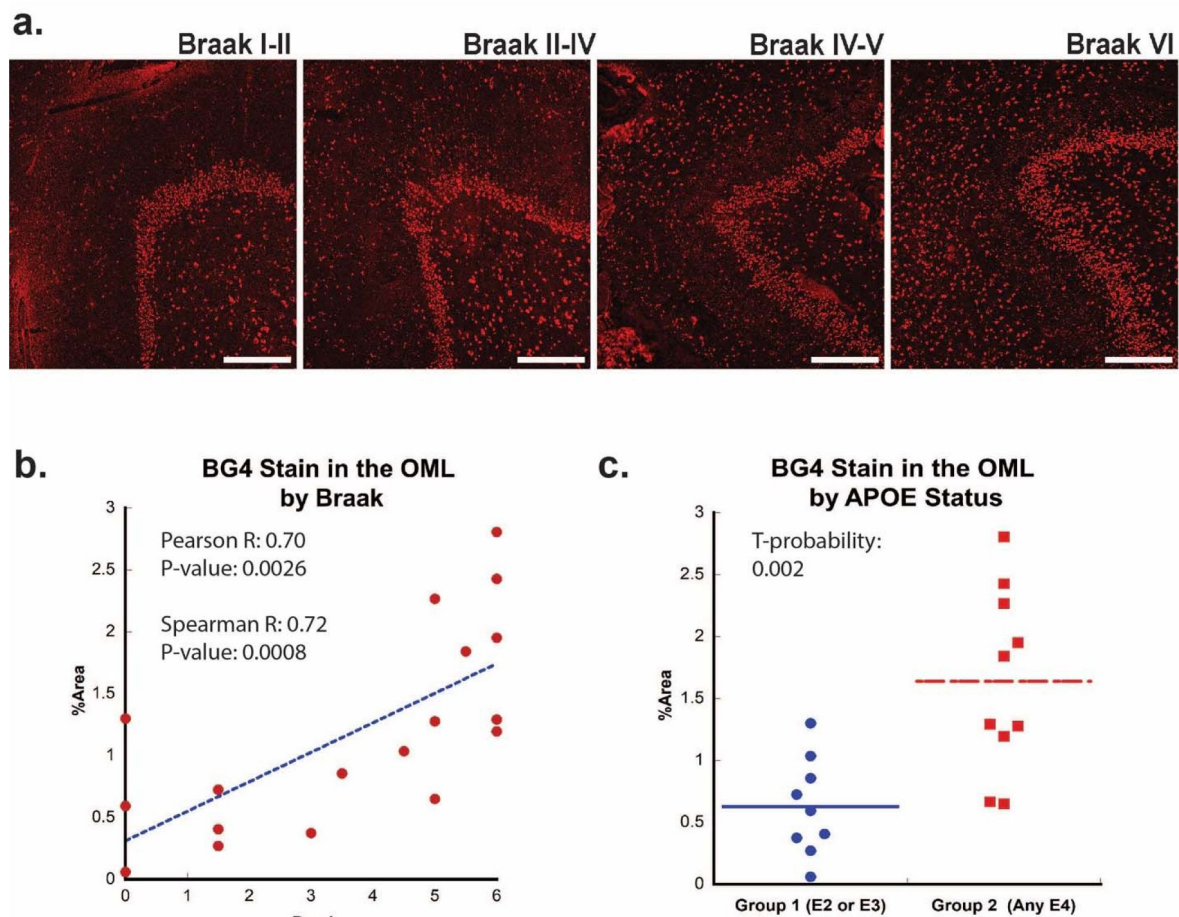


Fig 2.

a. Immunofluorescence showing BG4 staining (red) in brain sections of human hippocampus with increasing Braak stage (1-6) associated with AD severity (Case 16, 11, 10, and 8, respectively). All scale bars are 300 μ m. **b.** Correlation of percent area BG4 coverage in the outer molecular layer versus severity of Braak stage. **c.** Quantification of percent area BG4 coverage by population with E2/E3 and E3/E3 versus E3/E4 and E4/E4 APOE alleles.

showing that at least a large percentage of the observed BG4 signal arose from rG4s and not nuclear DNA G4s. This result also confirms that previously observed rG4 formation in ribosomes³³ also occurs in human brain tissue.

To further explore the formation of rG4s in the human hippocampus, we examined the staining pattern in different cell types based on specific antibody markers. Co-staining with markers for the most common brain cell types, rG4s were clearly abundant in neurons, oligodendrocytes, and astrocytes (**Fig. 3c**). rG4s exhibited no apparent colocalization with microglia (**Fig. 3c**). This pattern is especially noticeable in the double labeling with SOX10 (**Fig. 3c**), which clearly shows the nuclear oligodendrocyte marker SOX10³⁷ staining in proximity to the cytoplasmic BG4 immunostaining.

We co-stained BG4 with different p-tau epitope antibodies to determine whether pathological aggregation of tau proteins co-occurred with BG4 immunostaining (**Fig. 4**). Indeed, we found that BG4 immunostaining co-localized with hyper-phosphorylated tau immunostaining in neurons located in the CA1-4 regions of the hippocampus (**Fig. 4**). We used two different antibodies directed against the serine 396 site (S396) and the threonine 231 (T231) to investigate co-localization with different p-Tau epitopes. Quantifying the percentage of cells co-stained with BG4 and pTau revealed that 85% of the S396 p-tau positive cells also had positive BG4 staining, and 82% of the T231 p-tau positive cells also had positive BG4 staining.

4. Discussion

In this study, we demonstrated that rG4s form preferentially as a function of age and AD pathology in neurons and glial cells the human *postmortem* brain. While the granule cell layer in the dentate gyrus exhibited the most BG4 immunostaining overall, this was not an area most changed with aging and AD, as considerably greater changes were observed in the OML and the CA4 region of the hippocampus. rG4s were primarily found in the cytoplasm near the nucleus of neurons, astrocytes, and oligodendrocytes and were also found in neurons that contained regions of positive staining against two different epitopes of p-tau. The BG4 staining levels were also impacted by APOE status, with the presence of one or two APOE4 alleles associated with higher BG4 levels in the OML region. The OML is an interesting region of the hippocampus from the pathology standpoint, since this is the region of the hippocampus exhibiting the most prominent synaptic loss early in the disease process^{36, 38}. We have recently demonstrated a significant loss of presynaptic components of the OML in AD³⁸, strengthening the findings observed herein and suggesting a particular vulnerability of this region of the hippocampus to AD progressive pathology.

Combining these observations with those previously made on the reliance of rG4 structure on cellular stress and the roles of rG4s in protein oligomerization and tau aggregation *in vitro*, we can propose a model of how rG4s form and could perpetuate the effects of AD (**Fig. 4c**). As ROS and other protein stresses accumulate with age and according to APOE genotype, rG4s form. Although rG4s can rescue protein folding and could release proteins when unfolded at stress cessation³⁹, under persistent stress proteins RNA would not be released from the G-quadruplexes, leading to increased protein oligomerization and aggregation²⁶. This function has previously been suggested for RNA in biomolecular condensates more generally⁴⁰. The acceleration of protein oligomerization and aggregation by the rG4s could then contribute to a vicious cycle of increased aggregation, leading to synaptic loss as described recently for the OML³⁸. Recent *in vitro* work confirms an interaction between rG4 formation and p-Tau aggregation specifically⁴¹. *In vitro* experiments focused on the specific mechanisms for the rG4 influence on Tau phosphorylation and/or aggregation will be a future focus following these preliminary observations. Of note, the

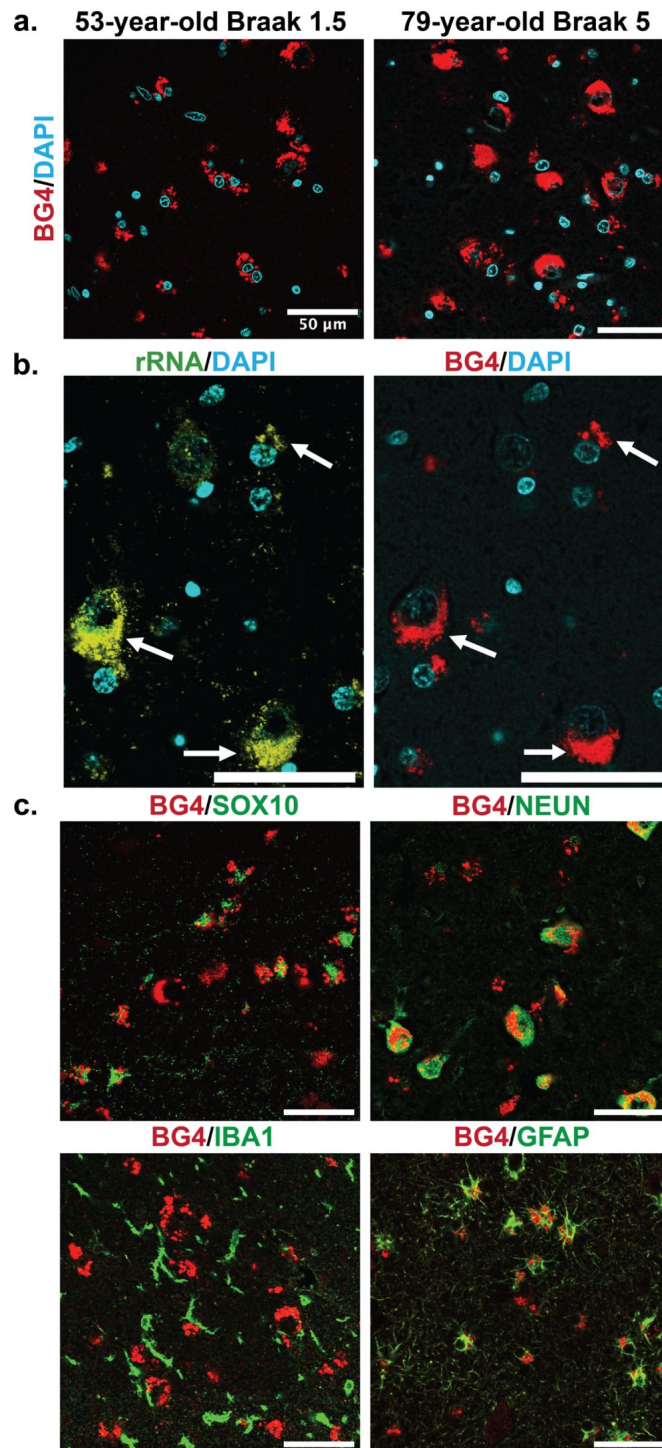


Fig 3.

Cellular types and localization of rG4s, all scale bars are 50 μm. **a**, Immunofluorescence showing BG4 staining patterns (red) in human hippocampal tissue of case 17, a 53-year-old individual with Braak stage 1.5, and case 4, a 79-year-old individual with Braak stage 5 AD pathology (right). Cell nuclei stained with DAPI (blue). **b**, Co-stain of rRNA (left, yellow green) and BG4 (right, red), in the hippocampal CA4 region, an 80-year old individual with early stage tauopathy, demonstrating significant but not complete overlap between BG4 and rRNA. **c**, BG4 staining (red) with cell type markers (green) of oligodendrocytes (SOX10, case 8), neurons (NEUN, case 4), microglia (IBA1, case 18) and astrocytes (GFAP, case 18). BG4 is most prevalent in oligodendrocytes, neurons, and astrocytes. BG4 does not strongly colocalize with microglia (IBA1).

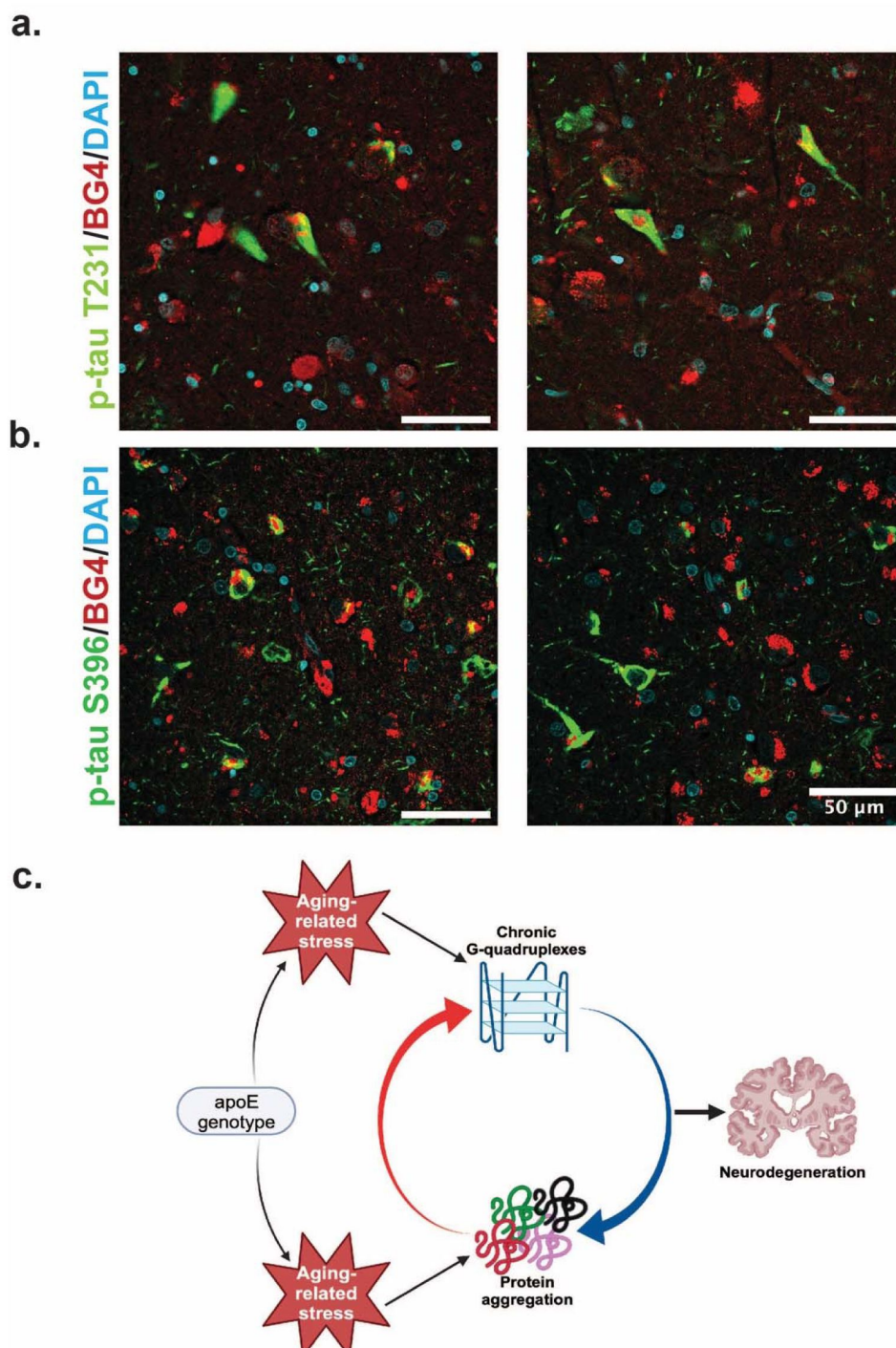


Fig 4.

Colocalization of rG4s with p-tau and neurodegeneration model.

a, Immunofluorescence of human hippocampus from 2 older AD individuals (case 8 and case 21) showing BG4 (red), DAPI (blue), and p-tau (green). **b**, Proposed model relating chronic rG4 formation and protein aggregation in aging and neurodegeneration.

previous studies showing that the fixation conditions used here strongly favor rG4 identification instead of DNA G4 identification were performed in cells, and so it is possible that in tissue some DNA staining could also occur, however this is not suggested by the co-stain with rRNA (**Fig. 3b** [↗](#)).

Of note, the accumulation of rG4s in neurodegeneration could have roles in addition to those of protein aggregation. RNA transport dysregulation has also been a common factor identified in neurodegeneration⁴² [↗](#), and it could be possible that the rG4 build-up observed here could be in part a product of this process or could lead to it. It has also been proposed that the RNA transport dysregulation and protein aggregation processes could be linked⁴³ [↗](#). This remains an area of needed future investigation.

This work raises the possibility of using rG4s as potential drug targets in AD, or given the high correlation with AD severity, as early AD biomarkers. Future work can focus on ameliorating AD symptoms through rG4 binding molecules or detecting brain-derived rG4s as disease biomarkers for diagnosis. Many different rG4-binding molecules have already been created and shown to bind in human cells⁴⁴ [↗](#), and provide a set of promising lead compounds for potential AD treatment.

Although here we have examined only AD, there are many other aging-related diseases featuring protein aggregation, including Parkinson's Disease, ALS, Fragile X syndrome, and Huntington's Disease⁴⁵ [↗](#). Notably, all of these diseases now have significant evidence showing rG4s as interactors with the aggregating proteins^{46–58} [↗](#), and in the case of Fragile X, a recent study demonstrated that rG4s could contribute to its aggregation and neurotoxicity⁵⁹ [↗](#). In addition, a recent manuscript by Raguseo et al., showed an increased prevalence of G4-structures in C9orf72 mutant human motor neurons in ALS/FTD when compared to healthy motor neurons, and direct involvement of the rG4 complex in the aggregation process⁶⁰ [↗](#). These recent findings illustrate the need to explore rG4 complexes and their role in neurodegeneration further. The results here combined with these previous studies suggest that the formation of rG4s could be a common mechanism of neurodegeneration.

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Additional information

Author Contributions

Idea generated by S.H., work conceptualized by S.H. and A.C.G. Experiments designed by S.H. and A.C.G., with assistance from all authors. Tissues collected and catalogued by E.D.H. Experiments performed by L.K., with assistance from H.S. and A.G. Primary writing performed by S.H., A.G.G., and L.K., with editing provided by all authors.

Authors declare no competing interests.

Additional files

Supplemental Materials [↗](#)

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Reviewer #1 (Public review):

This is an interesting manuscript where the authors systematically measure rG4 levels in brain samples at different ages of patients affected by AD. To the best of my knowledge this is the first time that BG4 staining is used in this context and the authors provide compelling evidence to show an association with BG4 staining and age or AD progression, which interestingly indicates that such RNA structure might play a role in regulating protein homeostasis as previously speculated. The methods used and the results reported seems robust and reproducible. There were two main things that needed addressing:

(1) Usually in BG4 staining experiments to ensure that the signal detected is genuinely due to rG4 an RNase treatment experiment is performed. This does not have to be extended to all the samples presented but having a couple of controls where the authors observe loss of staining upon RNase treatment will be key to ensure with confidence that rG4s are detected under the experimental conditions. This is particularly relevant for this brain tissue samples where BG4 staining has never been performed before.

(2) The authors have an association between rG4-formation and age/disease progression. They also observe distribution dependency of this, which is great. However, this is still an association which does not allow the model to be supported. This is not something that can be fixed with an easy experiment and it is what it is, but my point is that the narrative of the manuscript should be more fair and reflect the fact that, although interesting, what the authors are observing is a simple correlation. They should still go ahead and propose a model for it, but they should be more balanced in the conclusion and do not imply that this evidence is sufficient to demonstrate the proposed model. It is absolutely fine to refer to the literature and comment on the fact that similar observations have been reported and this is in line with those, but still this is not an ultimate demonstration.

Comments on current version:

The authors have now addressed my concerns.

<https://doi.org/10.7554/eLife.105446.1.sa2>

Reviewer #2 (Public review):

RNA guanine-rich G-quadruplexes (rG4s) are non-canonical higher order nucleic acid structures that can form under physiological conditions. Interestingly, cellular stress is positively correlated with rG4 induction.

In this study, the authors examined human hippocampal postmortem tissue for the formation of rG4s in aging and Alzheimer Disease (AD). rG4 immunostaining strongly increased in the hippocampus with both age and with AD severity. 21 cases were used in this study (age range 30-92).

This immunostaining co-localized with hyper-phosphorylated tau immunostaining in neurons. The BG4 staining levels were also impacted by APOE status. rG4 structure was previously found to drive tau aggregation. Based on these observations, the authors propose a model of neurodegeneration in which chronic rG4 formation drives proteostasis collapse.

This model is interesting, and would explain different observations (e.g., RNA is present in AD aggregates and rG4s can enhance protein oligomerization and tau aggregation).

Main issue from the previous round of review:

There is indeed a positive correlation between Braak stage severity and BG4 staining, but this correlation is relatively weak and borderline significant ($R = 0.52$, p value = 0.028). This is probably the main limitation of this study, which should be clearly acknowledged (together with a reminder that "correlation is not causality"). Related to this, here is no clear justification to exclude the four individuals in Fig 1d (without them R increases to 0.78). Please remove this statement. On the other hand, the difference based on APOE status is more striking.

Comments on current version:

The authors have made laudable efforts to address the criticisms I made in my evaluation of the original manuscript.

<https://doi.org/10.7554/eLife.105446.1.sa1>

Author response:

Reviewer #1 (Evidence, reproducibility and clarity):

This is an interesting manuscript where the authors systematically measure rG4 levels in brain samples at different ages of patients affected by AD. To the best of my knowledge this is the first time that BG4 staining is used in this context and the authors provide compelling evidence to show an association with BG4 staining and age or AD progression, which interestingly indicates that such RNA structure might play a role in regulating protein homeostasis as previously speculated. The methods used and the results reported seem robust and reproducible.

In terms of the conclusions, however, I think that there are 2 main things that need addressing prior to publication:

(1) Usually in BG4 staining experiments to ensure that the signal detected is genuinely due to rG4 an RNase treatment experiment is performed. This does not have to be extended to all the samples presented but having a couple of controls where the authors observe loss of staining upon RNase treatment will be key to ensure with confidence that rG4s are detected under the experimental conditions. This is particularly relevant for this brain tissue samples where BG4 staining has never been performed before.

With what is now known about RNA rG4s and the recent reconciliation of the controversy on rG4 formation (Kharel, Nature Communications 2023), this experiment is no longer strictly required for demonstration of rG4 formation. Despite this change, we did attempt this experiment at the reviewer's suggestion, but the controls were not successful, suggesting it may not be feasible with our fixing and staining conditions. That said, we agree that despite the G4 staining appearing primarily outside the nucleus, it would be helpful to have some direct indication of whether we were observing primarily RNA or DNA G4s, and so we performed an alternate experiment to determine this.

In our previous submission, we had performed ribosomal RNA staining (Figure S7), and the staining patterns were similar to that of BG4, especially the punctate pattern near the nuclei. Therefore, we directly asked whether the BG4 was largely binding to rRNA and have now shown the resulting co-stain in Figure 3b. These results show that at least a large amount of the BG4 staining does arise from rG4s in ribosomes. At high magnification, we observe that the BG4 stains a subset of the ribosomes, consistent with previous observations of high rG4 levels in ribosomes both in vitro and in cells (Mestre-Fos, 2019 J Mol Biol, Mestre-Fos 2019 PLoS One, Mestre-Fos 2020 J Biol Chem), but this had never been demonstrated in tissue. This experiment has therefore both answered the primary question of whether we are primarily observing rG4s, as well as provided more detailed information on the cellular sublocalization of rG4 formation, and provided the first evidence of rG4 formation on ribosomes in tissue.

(2) The authors have an association between rG4-formation and age/disease progression. They also observe distribution dependency of this, which is great. However, this is still an association which does not allow the model to be supported. This is not something that can be fixed with an easy experiment and it is what it is, but my point is that the narrative of the manuscript should be more fair and reflect the fact that, although interesting, what the authors are observing is a simple correlation. They should still go ahead and propose a model for it, but they should be more balanced in the conclusion and do not imply that this evidence is sufficient to demonstrate the proposed model. It is absolutely fine to refer to the literature and comment on the fact that similar observations have been reported and this is in line with those, but still this is not an ultimate demonstration.

We agree that these are correlative studies (of necessity when studying human tissue), but recent experiments have shown that rG4s affect the aggregation of Tau in vitro – and we have now better clarified this in the text itself. We have now also been more careful in drawing causative conclusions as shown in the revised text.

Minor point:

(3) rG4s themselves have been shown to generate aggregates in ALS models in the absence of any protein (Raguso et al. Nat Commun 2023). I think this is also important in the light of my comment on the model, could well be that these rG4s are causing aggregates themselves that act as nucleation point for the proteins as reported in the paper I mentioned. Providing a broader and more unbiased view of the current literature on the topic would be fair, rather than focusing on reports more in line with the model proposed.

We agree and have modified the discussion and added a broader context, including the Raguso report described above.

Reviewer #1 (Significance):

This is a significant novel study, as per my comments above. I believe that such a study will be of impact in the G4 and neurodegenerative fields. Providing that the authors can address the criticisms above, I strongly believe that this manuscript would be of value to the scientific community. The main strength is the novelty of the study (never done before) the main weakness is the lack of the RNase control at the moment and the slightly over interpretation of the findings (see comments above).

Reviewer #2 (Evidence, reproducibility and clarity):

RNA guanine-rich G-quadruplexes (rG4s) are non-canonical higher order nucleic acid structures that can form under physiological conditions. Interestingly, cellular stress is

positively correlated with rG4 induction. In this study, the authors examined human hippocampal postmortem tissue for the formation of rG4s in aging and Alzheimer Disease (AD). rG4 immunostaining strongly increased in the hippocampus with both age and with AD severity. 21 cases were used in this study (age range 30-92). This immunostaining co-localized with hyper-phosphorylated tau immunostaining in neurons. The BG4 staining levels were also impacted by APOE status. rG4 structure was previously found to drive tau aggregation. Based on these observations, the authors propose a model of neurodegeneration in which chronic rG4 formation drives proteostasis collapse.

This model is interesting, and would explain different observations (e.g., RNA is present in AD aggregates and rG4s can enhance protein oligomerization and tau aggregation).

Main issue:

There is indeed a positive correlation between Braak stage severity and BG4 staining, but this correlation is relatively weak and borderline significant ($R = 0.52$, p value = 0.028). This is probably the main limitation of this study, which should be clearly acknowledged (together with a reminder that "correlation is not causality").

We believe that we had not explained this clearly enough in the text (based on the reviewer's comment), as the correlation mentioned by the Reviewer was for the CA4 region only, and not the OML, which was substantially more correlated and statistically significant (Spearman $R = 0.72$, $p = 0.00086$). As a result, we believe this was a miscommunication that is rectified by the revised text:

"In the OML, plotting BG4 percent area versus Braak stage demonstrated a strong correlation (Spearman $R = 0.72$) with highly significantly increased BG4 staining with higher Braak stages ($p = 0.00086$) (Fig. 2b)."

Related to this, here is no clear justification to exclude the four individuals in Fig 1d (without them R increases to 0.78). Please remove this statement. On the other hand, the difference based on APOE status is more striking.

We did not mean to imply that deleting these outliers was correct, but merely were demonstrating that they were in fact outliers. To avoid this misinterpretation, we have now deleted the sentence in the Figure 1d caption mentioning the outliers.

Minor suggestions

- "BG4 immunostaining was in many cases localized in the cytoplasm near the nucleus in a punctate pattern". Define "many"

This is seen in nearly every cells and this is now altered in the text and is now identified as ribosomes containing rG4s using the rRNA antibody (Fig. 3b).

- Specify that MABE917 corresponds to the specific single-chain version of the BG4 antibody

Yes, this is correct, and this clarification has been added to the manuscript

- Define PMI, Braak, CERAD (add a list of acronyms or insert these definitions in Fig 1b legend)

These definitions have all been added when they first appear.

- Fig 3: scale bar legend missing (50 micrometers?)

This has been added, and the reviewer was correct that it was 50 micrometers.

- *Supplementary data Table 1: indicate target for all antibodies*

The target for each antibody has been added to supplementary Table 1.

- *Supplementary data Table 2: why give ages with different levels of precision? (e.g. 90.15 vs 63)*

We apologize for this oversight and have altered the ages to the same (whole years) in the figure.

- *Supplementary data Fig 1 X-axis legend: add "(nm)" after wavelength. Sequence can also be added in the legend. Why this one? Max/Min Wavelengths in the figure do not match indications in the experimental part. Not sure if that part is actually relevant for this study.*

The CD spectrum in Sup Fig 1 is the sequence that had previously been shown to aid in tau aggregation seeding, but had not been suspected by those authors to be a quadruplex. So we tested that here and showed it is a quadruplex, as described at the end of the introduction. We have added wording to the figure legend to clarify where its corresponding description in the main text can be found. We have also checked and corrected the wavelength and units.

- *Supplementary data Fig 7: Which ribosomal antibody was used?*

The details of this antibody have now been added to Supplementary Table 2 which lists all the antibodies used.

Reviewer #2 (Significance):

Provide a link between Alzheimer disease and RNA G-quadruplexes.

Reviewer #3 (Evidence, reproducibility and clarity):

This study investigated the formation of RNA G quadruplexes (rG4) in aging and AD in human hippocampal postmortem tissue. The rG4 immunostaining in the hippocampus increases strongly with age and with the severity of AD. Furthermore, rG4 is present in neurons with an accumulation of phosphorylated tau immunostaining.

Major comments

(1) The method used in this study is primarily immunostaining of BG4, and the results cannot be considered correct without additional data from more multifaceted analyses (biochemical analysis, RNA expression analysis, etc.).

We respectfully disagree with the Reviewer's assessment of the value of these experiments. The most relevant biochemical experiments at the cellular and molecular level showing the role of G4s in aggregation in general and Tau in particular have been done and are referenced in the text. The results here stand on their own and are highly novel and significant, as evaluated by both of the other reviewers. There has been no previous work demonstrating the presence of rG4s in human brain – either in controls or in patients with AD. AD is a complex condition that only occurs spontaneously in the human brain and no other species; because of this complexity, novel aspects are best first studied in human brain tissue using the methods employed here.

(2) Overall, the quality of the stained images is poor, and detailed quantitative analysis using further high quality data is essential to conclude the authors' conclusions.

We have again looked at our images and they are not poor quality -they are confocal images taken at recommended resolution of the confocal microscope. It is possible the poor quality came from pdf compression by the manuscript submission portal, which is beyond our control as they were uploaded at high resolution. These data were quantified by scientists who were blinded to the diagnosis of each case. The level of description on the detailed quantification is higher than we have observed in similar studies. We therefore disagree with the reviewer's conclusion.

Reviewer #3 (Significance):

Overall, this study is not a deeply analyzed study. In addition, the authors of this study need further understanding regarding G4.

It is also unclear why the reviewer believes that we do not have sufficient understanding of G4s, and would request that the reviewer instead provides specific comments regarding what is lacking in terms of knowledge on G4s, as we respectfully disagree with this judgement of our knowledge-base (see other G4 papers from the Horowitz lab, Begeman, 2020, Litberg 2023, Son, 2023 referenced below).

Litberg TJ, Sannapureddi RKR, Huang Z, Son A, Sathyamoorthy B, Horowitz S. Why are G-quadruplexes good at preventing protein aggregation? Jan;20(1):495-509. doi: 10.1080/15476286.2023.2228572. RNA Biol. (2023)

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Guzman BB, Son A, Litberg TJ, Huang Z, Dominguez , Horowitz S. Emerging Roles for G-Quadruplexes in Proteostasis FEBS J.doi: 10.1111/febs.16608. (2022)

Begeman A, Son A, Litberg TJ, Wroblewski TH, Gehring T, Huizar Cabral V, Bourne J, Xuan Z, Horowitz S. G-Quadruplexes Act as Sequence Dependent Protein Chaperones. EMBO Reports Sep 18;e49735. doi: 10.15252/embr.201949735. (2020)

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